

Qian ZhongJun 301541827 Aden Trathen 301451426

This project aims to develop a lightweight, real-time network security visualization tool inspired by SIEM functionalities using Observable. The tool will provide an interactive interface for visualizing network traffic anomalies, detecting potential denial-of-service (DoS) attacks, and monitoring suspicious activity in a controlled test environment.

The intended users for this system are cybersecurity professionals, SOC analysts, penetration testers, and security researchers who require an easy-to-deploy and interactive visualization tool for monitoring network security.

Users are expected to have:

Intermediate technical knowledge of networking

Basic familiarity with visualization tools and security data analysis.

Experience with SIEMs and IDS/IPS systems is helpful but not required.

We will generate our own dataset through a controlled cybersecurity test environment:

Red Team (Attacker): A Kali Linux machine will execute nmap scans, DoS attacks (SYN flood, ICMP flood)

Blue Team (Defender): A Linux-based machine will run Wireshark/tshark to capture real-time network traffic logs.

Inline IDS (Optional): Firewall rules or simple packet filtering scripts will be used to detect attack patterns.

Network Environment: A self-contained private LAN setup using a mobile hotspot or a VLAN configuration to isolate attack traffic from external networks.

To facilitate real-time visualization, we will:

Stream Wireshark logs via WebSockets into Observable, enabling real-time network monitoring.

Parse attack-specific features (e.g., packet spikes, SYN flood detection) to visualize network anomalies dynamically.

The proposed SIEM will feature multiple interactive views to explore network activity:

Real-Time Traffic Monitoring Dashboard

Live packet count visualization to detect abnormal traffic spikes.

Brushing & filtering features to zoom in on specific time windows.

Interactive Packet Drill-Down

Clickable data points that reveal detailed packet-level insights (source IP, destination IP, protocol, request type).